

- RAG Learning

2.4 Non-linear Graphs

Pearson Edexcel GCSE Mathematics 1MA1 — Algebra — spec refs A11–A16, R14 (graphs) — Papers 1 (non-calculator), 2 and 3 (calculator)

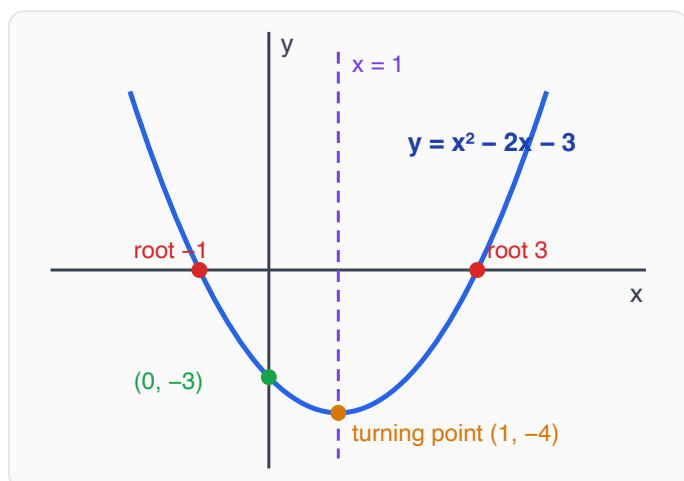
Both tiers cover the roots, intercepts and turning points of quadratic graphs, with roots found algebraically (A11); recognising and sketching linear, quadratic, simple cubic and reciprocal graphs (A12); plotting and interpreting real-life graphs, including distance, speed and acceleration (A14); and graphs of direct and inverse proportion (R14). Higher tier adds turning points by completing the square (A11), exponential and trigonometric graphs (A12 and A14), translations and reflections of graphs (A13), gradients of curves and areas under graphs (A15), and the circle $x^2 + y^2 = r^2$ with its tangents (A16).

2.4.1 Quadratic Graphs: Roots, Intercepts and Turning Points (A11, A12)

A **quadratic** function has the form $y = ax^2 + bx + c$ with $a \neq 0$. Its graph is a smooth curve called a **parabola**:

- if $a > 0$ it is U-shaped and its turning point is a **minimum**;
- if $a < 0$ it is n-shaped and its turning point is a **maximum**.

The curve is symmetrical about the vertical line through its turning point.



Feature	Method	For the curve shown, $y = x^2 - 2x - 3$
y-intercept	Substitute $x = 0$ (it is always the constant term)	$(0, -3)$
Roots	Where the curve meets the x -axis: solve $y = 0$	$x = -1$ and $x = 3$
Line of symmetry	Midway between the roots	$x = 1$
Turning point	Substitute the line of symmetry into the equation	$(1, -4)$

Plotting a Quadratic

Complete a table of values, plot every point and join them with a **smooth curve** — never with straight segments, and never with a flat bottom between two equal lowest points.

x	-4	-3	-2	-1	0	1	2	3
$y = x^2 + x - 6$	6	0	-4	-6	-6	-4	0	6

The values repeat symmetrically about $x = -\frac{1}{2}$, which is where the minimum lies (between the two values of -6).

Exam Tip: $(-3)^2 = 9$, not -9 . Put negative numbers in brackets on your calculator. A table of a quadratic that is not symmetrical almost always contains this error.

Deducing Roots Algebraically

The roots of $y = ax^2 + bx + c$ are the solutions of $ax^2 + bx + c = 0$. At Foundation tier you will normally factorise; then the line of symmetry lies halfway between the roots.

Worked Example: Find the roots and the turning point of $y = x^2 - 6x + 5$.

$$(x - 1)(x - 5) = 0 \Rightarrow x = 1 \text{ or } x = 5$$

$$\text{Line of symmetry } x = \frac{1 + 5}{2} = 3; \quad y = 3^2 - 6(3) + 5 = -4$$

Turning point **(3, -4)**, a minimum.

Worked Example: A stone is thrown upwards. Its height after t seconds is $h = 25t - 5t^2$ metres. When does it hit the ground, and what is its greatest height?

$$25t - 5t^2 = 0 \Rightarrow 5t(5 - t) = 0 \Rightarrow t = 0 \text{ or } t = 5: \text{ it lands after } \mathbf{5 \text{ seconds}}.$$

The maximum is halfway, at $t = 2.5$: $h = 25(2.5) - 5(2.5)^2 = 62.5 - 31.25 = \mathbf{31.25 \text{ m}}$.

Approximate Solutions from a Graph

To solve $f(x) = k$ from the graph of $y = f(x)$, draw the horizontal line $y = k$ and read the x -values where it crosses the curve. Readings from a graph are estimates, so give them to the accuracy of the grid.

Worked Example: Use the table for $y = x^2 + x - 6$ to solve $x^2 + x - 6 = 6$.

Draw $y = 6$. It meets the curve where $x = -4$ and $x = 3$.

2.4.2 Turning Points by Completing the Square (A11)

Higher

$$x^2 + bx + c = \left(x + \frac{b}{2}\right)^2 - \left(\frac{b}{2}\right)^2 + c$$

When $y = (x + p)^2 + q$, the turning point is $(-p, q)$. It is a minimum because $(x + p)^2$ can never be negative.

Worked Example: Find the turning point of $y = x^2 - 10x + 30$ and decide whether the graph crosses the x -axis.

$$x^2 - 10x + 30 = (x - 5)^2 - 25 + 30 = (x - 5)^2 + 5$$

Turning point **(5, 5)**. The lowest value of y is 5, which is above the x -axis, so there are **no real roots**.

Worked Example: Find the turning point of $y = x^2 + 4x - 3$.

$$(x + 2)^2 - 4 - 3 = (x + 2)^2 - 7, \text{ so the turning point is } (-2, -7).$$

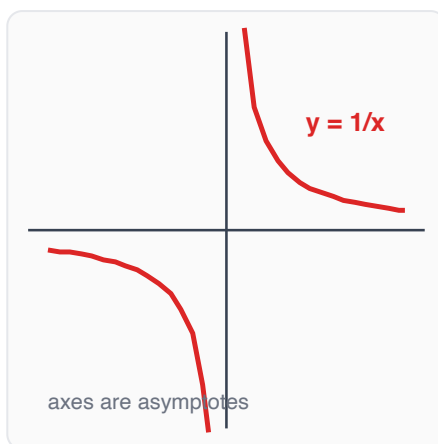
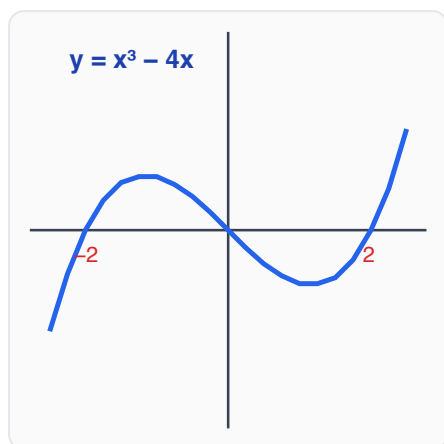
Worked Example: Find the turning point of $y = 3x^2 - 12x + 7$.

$$\text{Take out the 3 from the } x \text{ terms: } 3(x^2 - 4x) + 7 = 3[(x - 2)^2 - 4] + 7 = 3(x - 2)^2 - 5$$

Turning point **(2, -5)**.

Exam Tip: Only the x -coordinate changes sign: $(x - 5)^2 + 5$ gives **(5, 5)**, $(x + 2)^2 - 7$ gives **(-2, -7)**.

2.4.3 Recognising Cubic and Reciprocal Graphs (A12)



Type	General form	Shape and key features
Linear	$y = mx + c$	Straight line
Quadratic	$y = ax^2 + bx + c$	Parabola (U or n), one turning point, a line of symmetry
Cubic	$y = ax^3 + \dots$	"S"-shaped curve; meets the x -axis 1, 2 or 3 times. For $a > 0$ it rises from bottom-left to top-right; for $a < 0$ (e.g. $y = -x^3$) it falls from top-left to bottom-right.
Reciprocal	$y = \frac{k}{x}, x \neq 0$	Two separate branches that approach the axes but never touch them. For $k > 0$ the branches are in the 1st and 3rd quadrants.

An **asymptote** is a line that a curve approaches more and more closely without ever reaching. Both axes are asymptotes of $y = \frac{1}{x}$: x cannot be 0 (you cannot divide by zero), and y is never 0 however large x becomes.

Worked Example: Find where the graph of $y = x^3 - x^2 - 6x$ crosses the x -axis.

$x(x^2 - x - 6) = x(x - 3)(x + 2) = 0 \Rightarrow x = -2, 0, 3$. A positive cubic with three roots: it rises, turns down, turns up again.

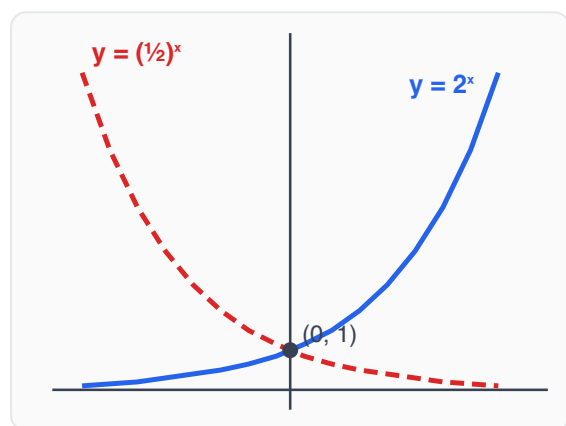
Direct and Inverse Proportion Graphs (R14)

- $y \propto x$, i.e. $y = kx$: a **straight line through the origin**. Doubling x doubles y .
- $y \propto \frac{1}{x}$, i.e. $y = \frac{k}{x}$: a **reciprocal curve**. Doubling x halves y , and the curve never meets the axes.

2.4.4 Exponential and Trigonometric Graphs (A12, A14)

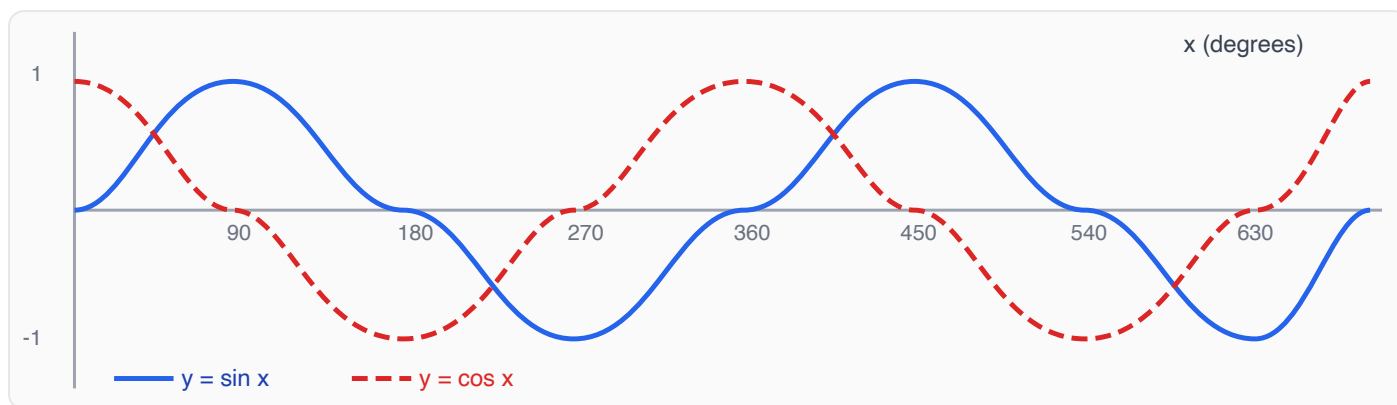
Higher

Exponential Graphs $y = k^x$



- For any positive k , the graph passes through $(0, 1)$ because $k^0 = 1$.
- y is always positive, so the curve stays above the x -axis, which is an asymptote.
- $k > 1$ gives exponential **growth**; $0 < k < 1$ gives exponential **decay**.
- In context, $V = A \times k^t$ starts at $(0, A)$. For example a car bought for £800 that loses 15% a year has $V = 800 \times 0.85^t$; after 2 years $V = 800 \times 0.7225 = £578$.

Trigonometric Graphs (angles in degrees)



Graph	Repeats every	Range	Key points for $0^\circ \leq x \leq 360^\circ$
$y = \sin x$	360°	$-1 \leq y \leq 1$	Zero at $0^\circ, 180^\circ, 360^\circ$; max 1 at 90° ; min -1 at 270°
$y = \cos x$	360°	$-1 \leq y \leq 1$	Max 1 at 0° and 360° ; zero at $90^\circ, 270^\circ$; min -1 at 180°
$y = \tan x$	180°	Any value	Zero at $0^\circ, 180^\circ, 360^\circ$; equals 1 at 45° ; asymptotes at 90° and 270°

Worked Example: Solve $\cos x = 0.5$ for $0^\circ \leq x \leq 360^\circ$.

A calculator gives $x = 60^\circ$. The cosine graph is symmetrical about $x = 180^\circ$, so the second solution is $360^\circ - 60^\circ = 300^\circ$.

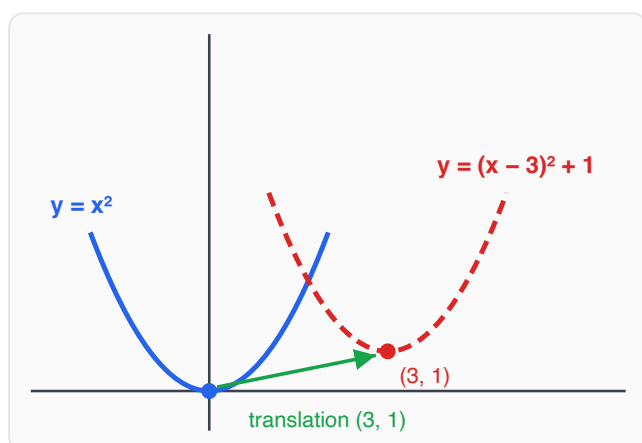
$x = 60^\circ$ and $x = 300^\circ$.

Exam Tip: Your calculator gives only one angle. Sketch the graph across the whole interval in the question, including negative angles or angles beyond 360° , and use its symmetry and period to find every solution.

2.4.5 Translations and Reflections of Graphs (A13)

Higher

New graph	Transformation of $y = f(x)$	Point (x, y) moves to
$y = f(x) + a$	Translation $\begin{pmatrix} 0 \\ a \end{pmatrix}$: up by a	$(x, y + a)$
$y = f(x + a)$	Translation $\begin{pmatrix} -a \\ 0 \end{pmatrix}$: left by a	$(x - a, y)$
$y = -f(x)$	Reflection in the x -axis	$(x, -y)$
$y = f(-x)$	Reflection in the y -axis	$(-x, y)$



Worked Example: The graph of $y = f(x)$ has a maximum point at $(3, 4)$ and crosses the x -axis at $(1, 0)$ and $(5, 0)$.

$y = f(x - 2)$: move 2 right $\Rightarrow$ maximum **$(5, 4)$** , roots **$(3, 0)$** and **$(7, 0)$** .

$y = -f(x)$: reflect in the x -axis $\Rightarrow$ it becomes a **minimum** at **$(3, -4)$** ; the roots stay where they are.

$y = f(x) + 2$: move 2 up $\Rightarrow$ maximum **$(3, 6)$** .

$y = f(-x)$: reflect in the y -axis $\Rightarrow$ maximum **$(-3, 4)$** , roots **$(-1, 0)$** and **$(-5, 0)$** .

Exam Tip: A change *inside* the bracket moves the graph horizontally and does the opposite of what the sign suggests: $f(x - 2)$ moves it 2 to the **right**. A change *outside* moves it

vertically in the direction you expect.

2.4.6 Real-Life and Kinematic Graphs (A14, R14)

Distance–time graph: the gradient is the **speed**. A horizontal section means the object is stopped; a steeper section means faster; a section sloping down means it is coming back towards the start.

Speed (velocity)–time graph: the gradient is the **acceleration**. A horizontal section means constant speed; a negative gradient means slowing down (deceleration).

Worked Example: A cyclist rides 18 km from home in 40 minutes, stops for 20 minutes, then rides back home in 1 hour. Work out her speed on each part of the journey.

Outward: $18 \div \frac{40}{60} = \mathbf{27 \text{ km/h}}$. Stopped: 0 km/h. Return: $18 \div 1 = \mathbf{18 \text{ km/h}}$.

Worked Example: A lorry's speed increases steadily from 6 m/s to 30 m/s in 12 seconds.

$$\text{Acceleration} = \frac{30 - 6}{12} = \mathbf{2 \text{ m/s}^2}.$$

Filling Containers

If water flows in at a constant rate, any part of a container with straight vertical sides gives a straight section on a depth–time graph. A narrow part fills quickly (steep line); a wide part fills slowly (shallow line). Sloping sides give a curve: a container that widens upwards gives a curve that gets less steep.

Other Contexts

Graphs of non-standard functions appear in contexts such as temperature over a day, the cost of electricity or the height of water in a harbour. Read values carefully from the scales and use the graph to estimate answers. Inverse proportion gives reciprocal curves: if n machines finish a job in

$t = \frac{60}{n}$ hours, the graph never reaches either axis. At Higher, growth and decay models give exponential curves.

2.4.7 Gradients of Curves and Areas Under Graphs (A15)

Higher

Gradient of a Curve

- The **gradient at a point** on a curve is the gradient of the **tangent** drawn at that point (the instantaneous rate of change).
- The gradient of a **chord** joining two points gives the **average** rate of change between them.
- On a distance–time curve the tangent gradient is the speed at that moment; on a velocity–time curve it is the acceleration.

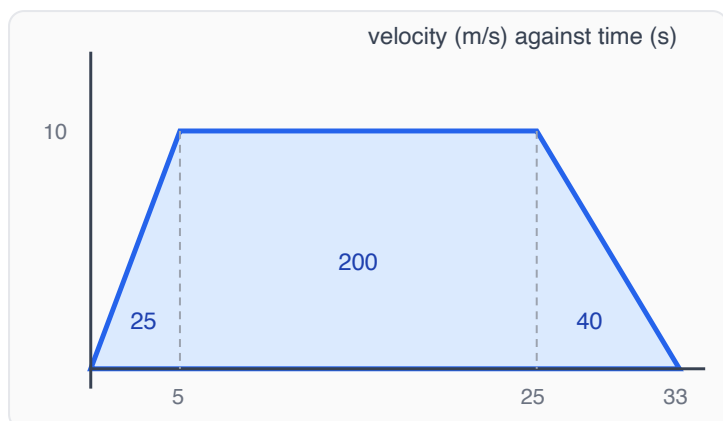
Worked Example: A tangent is drawn to a velocity–time curve at $t = 4$. It passes through $(2, 5)$ and $(8, 23)$. Estimate the acceleration at $t = 4$.

$$\text{Gradient} = \frac{23 - 5}{8 - 2} = \frac{18}{6} = \mathbf{3 \text{ m/s}^2}.$$

Area Under a Graph

The area under a velocity–time graph is the **distance travelled**. Split straight-line graphs into triangles, rectangles and trapezia. For a curve, estimate the area using **strips of equal width** treated as trapezia.

$$\text{Area of one trapezium strip} = \frac{1}{2} \times \text{width} \times (\text{left height} + \text{right height})$$



Worked Example: A tram accelerates from rest to 10 m/s in 5 s, travels at 10 m/s for 20 s, then slows steadily to rest in 8 s. Work out the total distance travelled.

$$\frac{1}{2} \times 5 \times 10 + 20 \times 10 + \frac{1}{2} \times 8 \times 10 = 25 + 200 + 40 = \mathbf{265 \text{ m}}$$

Worked Example: A velocity–time curve passes through (0, 2), (3, 8), (6, 11) and (9, 12). Use three strips to estimate the distance travelled in the first 9 seconds.

$$\frac{1}{2}(3)(2 + 8) + \frac{1}{2}(3)(8 + 11) + \frac{1}{2}(3)(11 + 12) = 15 + 28.5 + 34.5 = \mathbf{78 \text{ m}}$$

The curve gets less steep, so it lies above the tops of the trapezia: the estimate is an **underestimate**.

Exam Tip: Units come from the axes: gradient of a velocity–time graph is $(\text{m/s}) \div \text{s} = \text{m/s}^2$, area is $(\text{m/s}) \times \text{s} = \text{m}$. In a financial graph the gradient might be £ per month.

2.4.8 The Equation of a Circle and Tangents (A16)

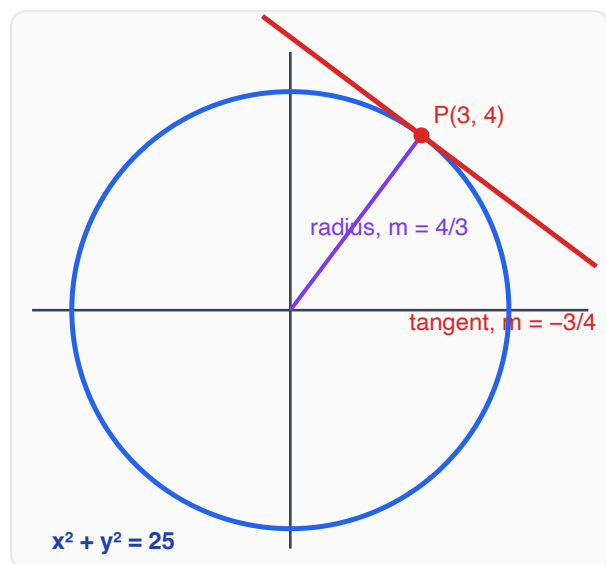
Higher

The circle with centre the origin (0, 0) and radius r has equation

$$x^2 + y^2 = r^2$$

For example, $x^2 + y^2 = 121$ has radius 11. A point lies on the circle exactly when its coordinates satisfy the equation.

The diagram shows $x^2 + y^2 = 25$ with the tangent at $(3, 4)$: the tangent is at right angles to the radius.



Tangent to $x^2 + y^2 = r^2$ at the point P :

1. Gradient of the radius $OP = \frac{y_P}{x_P}$.
2. The tangent is perpendicular to OP : its gradient is the negative reciprocal.
3. Substitute P into $y = mx + c$ to find c .

Worked Example: Find an equation of the tangent to $x^2 + y^2 = 29$ at the point $(5, 2)$.

Check: $25 + 4 = 29 \checkmark$. Gradient of radius $= \frac{2}{5}$, so the tangent gradient is $-\frac{5}{2}$.

$$2 = -\frac{5}{2}(5) + c \Rightarrow c = 2 + 12.5 = 14.5$$

$$y = -\frac{5}{2}x + 14.5, \text{ i.e. } 5x + 2y = 29$$

Worked Example: Find an equation of the tangent to $x^2 + y^2 = 17$ at $(-1, 4)$.

Radius gradient = $\frac{4}{-1} = -4$, so the tangent gradient is $\frac{1}{4}$.

$$4 = \frac{1}{4}(-1) + c \Rightarrow c = \frac{17}{4} \Rightarrow y = \frac{1}{4}x + \frac{17}{4}, \text{ i.e. } 4y = x + 17$$

Exam Tip: The number on the right is r^2 , not r : $x^2 + y^2 = 72$ has radius $\sqrt{72} = 6\sqrt{2}$. To find the equation of a circle through a given point, e.g. $(-5, 12)$, work out $r^2 = 25 + 144 = 169$, giving $x^2 + y^2 = 169$.

Quick Reference Summary

Idea	Key fact
Roots and turning point	Roots: solve $y = 0$; turning point on the line of symmetry midway between the roots
Completing the square Higher	$(x + p)^2 + q \Rightarrow$ turning point $(-p, q)$
Graph shapes	Quadratic U or n; cubic S; reciprocal two branches with asymptotes
Proportion	$y = kx$ straight line through O ; $y = \frac{k}{x}$ reciprocal curve
Exponential Higher	$y = k^x$ passes through $(0, 1)$
Trig graphs Higher	sin and cos repeat every 360° ; tan every 180°
Transformations Higher	$f(x) + a$ up a ; $f(x + a)$ left a ; $-f(x)$ and $f(-x)$ reflect
Kinematics	D–T gradient = speed; V–T gradient = acceleration
Gradients and areas Higher	Tangent = instantaneous rate; chord = average rate; area under V–T = distance
Circle Higher	$x^2 + y^2 = r^2$; tangent $\perp$ radius

